



UNIVERSIDAD TECNOLÓGICA EL RETOÑO

Major:

Information Technologies and Digital Innovation Engineering

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Subject:

Data Science

Group:

LITD9A

Activity 1. Data Opportunity Mapping

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1. Activity Type

Team Work

2. Activity Context and Continuity

In the previous activities in Disruptive Technologies and Integrative Project III, each team identified a real-world challenge, analyzed stakeholders, explored needs, investigated innovation opportunities, and defined a preliminary solution concept.

Before developing analytical models, artificial intelligence solutions, or predictive systems, organizations must understand what data are available, what information is needed, how data can be collected, and which variables may contribute to solving the selected problem.

This activity introduces students to the fundamental concepts of Data Science by focusing on the relationship between problems, data, information, and decision-making. Teams will analyze their selected challenge from a data perspective, identify potential sources of information, evaluate data availability and quality, and define an initial set of variables that could support future analytical and AI-based solutions.

The outputs generated in this activity will serve as the foundation for future activities involving data acquisition, data engineering, exploratory analysis, machine learning, and artificial intelligence.

3. Objective

To identify, evaluate, and map the data opportunities associated with a selected challenge through problem understanding, source identification, data assessment, and preliminary variable selection.

4. Expected Learning Outcomes

Knowledge

Students will be able to:

- Explain the role of data in problem-solving.
- Describe the Data Science lifecycle.
- Differentiate data, information, and knowledge.
- Identify structured and unstructured data sources.
- Explain the characteristics of Big Data.
- Recognize variables relevant to a specific problem.

Skills

Students will be able to:

- Analyze problems from a data perspective.
- Identify potential data sources.



- Evaluate data availability and quality.
- Select relevant variables.
- Organize information requirements.
- Create a data opportunity map.

Professional Attitudes

Students will develop:

- Analytical thinking.
- Evidence-based decision making.
- Critical evaluation skills.
- Curiosity and exploration.
- Data-driven mindset.
- Collaborative problem solving.

5. General Instructions

- Work with the same challenge selected during the previous activities.
- Use reliable information sources.
- Document every assumption and decision.
- Support all findings with evidence.
- Include diagrams, tables, and visual representations whenever appropriate.
- Focus on understanding what data are necessary before discussing technological implementation.

PHASE 1: UNDERSTANDING THE PROBLEM FROM A DATA PERSPECTIVE

Step 1 – Problem Reframing Through Data

Review your selected challenge.

Answer:

- What decision needs to be supported?
- What information is currently missing?
- How could data help address the problem?
- What outcomes would stakeholders like to predict, monitor, optimize, or understand?

Expected Output

Data-Oriented Problem Statement.

Step 2 – Decision-Making Analysis

Identify the key decisions that users and stakeholders must make.

Expected Output

Decision	Decision Maker	Information Required
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Step 3 – Information Needs Assessment

Determine the information required to support each decision.

Expected Output

Information Needs Matrix.

PHASE 2: DATA SOURCE EXPLORATION

Step 4 – Data Source Identification

Identify possible sources of information related to the challenge.

Consider:

- Internal organizational data
- Public datasets
- Government databases
- Sensors and IoT devices
- Social media
- Surveys
- Mobile applications
- Scientific repositories
- Open Data portals

Expected Output

Data Source Inventory.

Step 5 – Source Classification

Classify each data source.

Expected Output

Data Source	Internal/External	Structured/Unstructured	Potential Use



Step 6 – Data Availability Analysis

Evaluate the accessibility of each source.

Expected Output

Data Source	Available	Restricted	Estimated Difficulty

PHASE 3: BIG DATA OPPORTUNITY ANALYSIS

Step 7 – Big Data Characteristics

Analyze how the challenge relates to Big Data concepts.

Evaluate:

- Volume
- Velocity
- Variety
- Veracity
- Value

Expected Output

5Vs Analysis Table.

Step 8 – Data Generation Ecosystem

Identify how data are generated within the selected domain.

Expected Output

Data Ecosystem Diagram.

PHASE 4: VARIABLE IDENTIFICATION

Step 9 – Potential Variable Discovery

Brainstorm all possible variables related to the challenge.

Consider:

- Demographic variables
- Behavioral variables
- Geographic variables
- Temporal variables
- Operational variables



- Environmental variables

Expected Output

Variable Inventory.

Step 10 – Variable Classification

Classify each variable.

Expected Output

Variable	Independent	Dependent	Categorical	Numerical

Step 11 – Variable Relevance Assessment

Analyze the importance of each variable.

Expected Output

Variable Prioritization Matrix.

PHASE 5: DATA OPPORTUNITY MAPPING

Step 12 – Data Opportunity Map

Develop a visual map connecting:

- Problem
- Stakeholders
- Decisions
- Information Needs
- Data Sources
- Variables

Expected Output

Data Opportunity Map.

Step 13 – Gap Analysis

Identify missing information and unavailable data.

Expected Output

Gap Analysis Report.

Step 14 – Risk Analysis

Evaluate potential risks related to data acquisition.

Examples:

- Missing data
- Incomplete records
- Privacy concerns
- Data quality issues
- Data ownership restrictions

Expected Output

Data Risk Matrix.

PHASE 6: STRATEGIC RECOMMENDATIONS

Step 15 – Data Collection Recommendations

Recommend the most valuable sources for future project phases.

Expected Output

Data Collection Strategy.

Step 16 – Initial Dataset Proposal

Define the dataset that could be built for future analyses.

Include:

- Variables
- Sources
- Collection frequency
- Update frequency

Expected Output

Preliminary Dataset Design.

Step 17 – Reflection

Discuss:

- What information is currently available?
- What information is missing?
- What variables appear most important?
- What challenges could arise during data collection?

Expected Output

Reflection Report.

6. Deliverables



- Data-Oriented Problem Statement
- Decision-Making Analysis
- Information Needs Matrix
- Data Source Inventory
- Data Source Classification Table
- Data Availability Analysis
- Big Data 5Vs Analysis
- Data Ecosystem Diagram
- Variable Inventory
- Variable Classification Table
- Variable Prioritization Matrix
- Data Opportunity Map
- Gap Analysis Report
- Data Risk Matrix
- Data Collection Strategy
- Preliminary Dataset Design
- Reflection Report

7. Evaluation Criteria

Criterion	Weight
Problem Understanding from a Data Perspective	15%
Data Source Identification	15%
Big Data Analysis	10%
Variable Identification and Classification	20%
Data Opportunity Map	15%
Gap and Risk Analysis	10%
Data Collection Strategy	10%
Reflection and Conclusions	5%

8. References

Use **APA 7th Edition** format.

Include:

- Scientific articles
- Government datasets
- Open data portals
- Technical documentation
- Industry reports
- Books
- Reliable online resources



1. Introduction

Wildfire management across Mexico is confronting an acute operational crisis rooted in a systemic technical lag: national and state civil protection systems operate almost exclusively under reactive suppression frameworks. Current protocols rely heavily on thermal anomaly detections from orbital radiometers (e.g., NASA FIRMS VIIRS/MODIS) or macro-scale meteorological indices (e.g., the Canadian Forest Fire Weather Index / ISPPF). While valuable for post-combustion tracking, these mechanisms notify command centers only after ignition has occurred and flame fronts have expanded across critical terrain.

Concurrently, Mexico's complex mountainous topographies suffer from a severe scarcity of in-situ meteorological instrumentation, with fewer than one automated weather station per 2,500 km² in rugged forest biomes. This leaves municipal civil protection units, ejidal authorities, and volunteer brigades operating in a condition of environmental data blindness. Furthermore, the operational frontline—composed of rural ejidatarios and municipal brigades—navigates ravines and high-elevation cordilleras completely cut off from cellular telecommunications and public electric infrastructure.

The purpose of this Data Opportunity Mapping analysis is to structurally reframe the wildfire challenge from an emergency response crisis into a predictive data science and Big Data engineering problem. Transitioning from reactive containment to pre-ignition mitigation 48 to 72 hours in advance requires capturing, synchronizing, downscaling, and analyzing high-velocity, high-volume Earth Observation (EO) streams and global atmospheric reanalyses. This document establishes the foundational data architecture, catalogs available open-access environmental sources, evaluates Big Data complexities (Volume, Velocity, Variety, Veracity, Value), classifies predictive variables, maps operational data dependencies, and outlines a resilient data ingestion strategy designed to support predictive gradient-boosted modeling (XGBoost) and tactical offline delivery (PWA).

2. Problem Understanding

What decision needs to be supported?

Operational and preventive decisions: (1) preemptive resource prepositioning (crews, water tenders, backfire equipment), (2) proactive closure or surveillance of high-risk forest zones, and (3) real-time tactical routing and safety perimeter management for frontline volunteer and ejidal brigades operating in zero-connectivity terrain.

What information is currently missing?

Hyper-local (10-to-20-meter resolution) live fuel moisture content (LFMC) desiccation dynamics, micro-topographic wind acceleration vectors, downscaled continuous vapor pressure deficit (VPD) curves, and spatiotemporal ignition probability surfaces forecasted 48 to 72 hours prior to initial combustion.



How could data help address the problem?

By algorithmically coupling multi-spectral optical reflectance and radar backscatter with thermodynamic atmospheric reanalysis and digital elevation models, predictive machine learning models can detect sub-surface and canopy-level vegetative desiccation trends before combustion occurs.

What outcomes would stakeholders like to predict, monitor, optimize, or understand?

- *Predict*: Binary ignition probability and potential rate-of-spread severity within discrete geospatial grid cells ($20\text{ m} \times 20\text{ m}$) over a 48–72h horizon.
- *Monitor*: Live canopy water stress, surface soil moisture depletion, and local thermodynamic evaporative demand.
- *Optimize*: Allocation of municipal suppression assets and pre-fire patrol routes across vulnerable ejidal borders.
- *Understand*: The primary environmental drivers triggering extreme fire susceptibility via explainable feature attributions (SHAP values).

DECISION-MAKING ANALYSIS: STAKEHOLDERS & INFORMATION REQUIREMENTS

DECISION	DECISION MAKER	INFORMATION REQUIRED	CADENCE / URGENCY
Preemptive Crew Deployment & Resource Prepositioning DEC-01 • Tactical staging	State & Municipal Civil Protection, CONAFOR Regional Operational Coordinators & Command Chiefs	• 48–72h predicted ignition probability maps • Critical fuel desiccation zones (NDMI/LFMC) • Road network topology & response accessibility	DAILY (06:00 HRS) Updated every 24h
Controlled Burn Approvals & Forest Access Restrictions DEC-02 • Preventive moratoria	Municipal Forestry Authorities, Ejidal Commissariats & Local Fire Marshals	• Real-time atmospheric drying demand (VPD) • Dead fuel moisture depletion indices • 72h wind velocity vectors & gust forecasts	BI-WEEKLY / PEAK Daily during estiaje
Field Tactical Routing & Safe Evacuation Escapes DEC-03 • Fireline safety	Frontline Forest Brigades, Rural Volunteer Ejidatarios & Rapid Initial Attack Crews	• Offline vector-tiled localized risk surfaces • Micro-topographic slope/aspect chimney zones • Real-time localized wind channels & escape paths	REAL-TIME FIELD Offline PWA Runtime
Post-Fire Perimeter Assessment & Burn Severity Mapping DEC-04 • Damage verification	CONAFOR Environmental Analysts, Post-Incident Technical Evaluators & Civil Protection Restoration Teams	• Differenced Normalized Burn Ratio (dNBR) • Pre- and post-fire multispectral Sentinel-2 imagery • Total burned area polygons & biomass loss	< 48H POST-FIRE Post-containment cycle

Pipeline Traceability: All decision outputs feed directly into the offline-first Service Worker cache and PWA spatial vector layer.



INFORMATION NEEDS ASSESSMENT: MULTI-SOURCE ENVIRONMENTAL REQUIREMENTS

Phase 1 (Step 3) — Technical mapping of biophysical, thermodynamic, and spatial data inputs to predictive requirements

INFORMATION NEED	REQUIRED GRANULARITY	TARGET ENTITIES / FEATURES	DATA SOURCE	FUNCTIONAL OUTPUT
Canopy Water Stress & Live Fuel Desiccation Biophysical vegetative health	10–20m Spatial Res. 5-day revisit cadence <6h cloud processing latency	• NDMI: (B08 - B11) / (B08 + B11) • NDWI: (B08 - B12) / (B08 + B12) • SAVI: Canopy soil correction	SENTINEL-2 MSI L2A Bottom-Of-Atmosphere	LFMC Proxy Layer Dynamic live fuel flammability index
Micro-Atmospheric Evaporative Demand Thermodynamic drying force	Hourly temporal cadence 0.1° (~9 km) downscaled to 30m via DEM lapse rates	• 2m Air Temp (T _{air}) & Dewpoint • Vapor Pressure Deficit (VPD in kPa) • 10m U/V horizontal wind vectors	ECMWF ERA5-LAND Hourly CDS API feed	Atmospheric Drying Dynamic evaporative stress surface
Terrain Slope, Aspect & Solar Exposure Physical propagation physics	30m Spatial Resolution Static baseline layer Zero operational latency	• Slope (degrees, 0°–90°) • Aspect (solar irradiation radians) • Topographic Wetness Index (TWI)	COPERNICUS GLO-30 Global DEM / SRTM	Terrain Multiplier Slope acceleration & canyon wind funneling
Surface & Sub-Surface Soil Moisture Content Understory fuelbed drought	Daily / Multi-layer 0–7cm (Layer 1) 7–28cm (Layer 2)	• Volumetric soil water (m³/m³) • Synthetic Aperture Radar cross-ratio • Dielectric surface roughness	ERA5 & SENTINEL-1 Reanalysis + C-SAR	Understory Drought Dead surface fuelbed flammability threshold
Historical Wildfire Ignition Truth Labels Supervised model training	Historical polygon series 375m VIIRS spatial points Temporal match to 48-72h	• GPS fire perimeter boundaries • Thermal anomaly coordinates (Lat/Lon) • Ignition timestamp cross-validation	CONAFOR & FIRMS Federal logs + VIIRS	Binary Target Labels Y ∈ {0, 1} Ignition for XGBoost training

3. Data Source Exploration

Data Source Identification & Inventory

Data sources span satellite constellations, planetary-scale reanalysis models, government historical registries, and terrain elevation grids.

1. **Copernicus Sentinel-2 (MSI):** High-resolution multispectral imagery providing 13 optical/infrared bands for vegetative biophysical parameter estimation.
2. **ECMWF ERA5-Land Reanalysis:** High-density global atmospheric and land-surface reanalysis dataset delivering hourly meteorological and thermodynamic records.
3. **NASA FIRMS (MODIS / VIIRS):** Satellite-derived active fire hotspots and thermal anomalies detected via mid-infrared and thermal infrared sensors.
4. **CONAFOR Historical Wildfire Databases:** Official Mexican federal fire perimeters, ignition dates, causes, combat durations, and timber/shrubland damage metrics.
5. **Copernicus GLO-30 DEM / INEGI CEM:** High-accuracy Digital Elevation Models capturing ground topography, slope inclination, and aspect.
6. **OpenStreetMap (OSM) Road & Urban Layers:** Geospatial vector databases mapping road networks, powerlines, agricultural parcel boundaries, and human infrastructure.



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DATA SOURCE CLASSIFICATION: STRUCTURAL TAXONOMY & TARGET UTILIZATION

Phase 2 (Step 5) — Categorization of planetary, atmospheric, and federal data inputs for TERRA-FIRE

DATA SOURCE	ORIGIN	STRUCTURE TYPE	FORMAT / PROTOCOL	POTENTIAL USE IN TERRA-FIRE
Sentinel-2 MSI ESA / Copernicus 10–20m multispectral	EXTERNAL	SEMI-STRUCTURED Georeferenced Rasters	COG / GeoTIFF STAC API, Google Earth Engine	Derivation of biophysical fuel moisture indices: NDMI, NBR, NDVI, and SAVI
Sentinel-1 C-SAR ESA / Copernicus C-band radar backscatter	EXTERNAL	SEMI-STRUCTURED Radar Polarimetric Arrays	SAFE / GRD GeoTIFF Copernicus Data Space Ecosystem	Cloud-penetrating moisture tracking via dual-polarization cross-ratio (VH/VV)
ERA5-Land Reanalysis ECMWF / Copernicus Hourly climate grid (~9 km)	EXTERNAL	STRUCTURED Multidimensional Arrays	NetCDF-4 / GRIB2 CDS API / Python Open-Meteo	Calculation of continuous VPD (kPa), ambient temp, wind vectors, and soil moisture
Copernicus GLO-30 DEM ESA / OpenTopography 30m digital elevation grid	EXTERNAL	STRUCTURED Raster Elevation Grids	Cloud-Optimized GeoTIFF AWS S3 / Planetary Computer	Topographic slope/aspect calculation and adiabatic lapse-rate atmospheric downscaling
CONAFOR Fire Logs Federal Forestry (Mexico) Historical ignition registries	EXTERNAL	STRUCTURED Relational / Geospatial Vectors	CSV / Shapefile Portal Datos Abiertos MX	Supervised ground-truth training targets for binary ignition classification (0/1)
NASA FIRMS (VIIRS) NASA Earthdata 375m active fire detections	EXTERNAL	SEMI-STRUCTURED Point Feature Coordinates	REST API / GeoJSON Near-Real-Time Web Service	Spatio-temporal validation and ignition timestamp cross-referencing against logs
CONABIO / OSM CONABIO / OpenStreetMap Land cover and road lines	EXTERNAL	STRUCTURED Vector Polygons & Polylines	GeoPackage / Shapefile Overpass API / Direct Download	Fuel type stratification and calculation of proximity to roads and crop edges

DATA AVAILABILITY ANALYSIS: ACCESSIBILITY, LICENSING & INGESTION FRICTION

Phase 2 (Step 6) — Technical feasibility audit of external planetary streams, reanalyses, and federal registries

DATA SOURCE	AVAILABILITY STATUS	LICENSING / RESTRICTION	ESTIMATED DIFFICULTY	TECHNICAL ASSESSMENT & BOTTLENECKS
Sentinel-2 MSI ESA / Copernicus Optical (10-20m)	FULLY AVAILABLE	Open Access Mandate Free reuse (CC BY-SA 3.0 IGO)	MEDIUM DIFFICULTY	Automated STAC/GEE ingestion; cloud masking and heavy multiband GeoTIFF payload overhead
Sentinel-1 C-SAR ESA / Copernicus Radar Backscatter	FULLY AVAILABLE	Copernicus Data Policy Full, open, and unrestricted	MEDIUM-HIGH DIFF.	Requires thermal noise removal, terrain radiometric calibration, and speckle filtering
ERA5-Land Climate ECMWF / CDS Hourly Reanalysis	FULLY AVAILABLE	ECMWF Open Data Free access via CDS API	MEDIUM DIFFICULTY	Coarse 9 km grid demands downscaling via DEM lapse rates and NetCDF multidimensional parsing
Copernicus GLO-30 DEM ESA / OpenTopography 30m Topography	FULLY AVAILABLE	Public Domain / Open Access Unrestricted scientific reuse	LOW DIFFICULTY	Static reference surface; single-cycle download and local slope/aspect extraction in GIS/GDAL
CONAFOR Fire Logs Federal Forestry (Mexico) Historical Fire Perimeters	CONDITIONAL ACCESS	Datos Abiertos México Subject to annual portal release	MEDIUM DIFFICULTY	Coordinate truncations, missing ignition timestamps, and unstandardized shapefiles requiring cleaning
NASA FIRMS (VIIRS) NASA Earthdata 375m Thermal Hotspots	FULLY AVAILABLE	NASA Open Data Policy Unrestricted real-time REST API	LOW DIFFICULTY	Direct HTTP GET requests to FIRMS endpoints; straightforward spatial CSV/GeoJSON parsing
OSM & INEGI Vectors Community / INEGI Roads & Crop Boundaries	FULLY AVAILABLE	ODbL / Libre Uso INEGI Open database / attribution	LOW DIFFICULTY	High availability via Overpass API; Euclidean distance transforms executed locally in GIS/Python



4. Big Data Opportunity Analysis

BIG DATA 5Vs ANALYSIS: TERRA-FIRE DATA OPPORTUNITY & INFRASTRUCTURE

Phase 3 (Step 7) — Evaluation of Volume, Velocity, Variety, Veracity, and Value in predictive wildfire intelligence

BIG DATA "V"	DIMENSION ANALYSIS IN TERRA-FIRE	IMPACT ON ARCHITECTURE & ANALYTICS STRATEGY
VOLUME Terabytes of Rasters	Multi-temporal satellite imagery & climate reanalysis grids 13-band Sentinel-2 granules (~500 MB/granule at 10-20m) combined with multi-year hourly ERA5-Land reanalysis cubes across Mexico.	- Cloud object storage (AWS S3 / GEE) replaces local disks - Bounding box extraction & Cloud-Optimized GeoTIFFs (COGs) - Extraction of lightweight tabular training samples in Parquet
VELOCITY 5-Day to Hourly Feeds	Dynamic environmental shifts & recurring orbital updates Sentinel-2 revisits biomes every 5 days, while ERA5-Land produces continuous hourly updates. Predictions require daily 48-72h batches.	- Scheduled cron workflows (Apache Airflow / Serverless) - Incremental ETL triggers upon new image availability - Batch inference routines completed during low-load periods
VARIETY Multi-Format Pipeline	Heterogeneous spatial, multidimensional, & tabular data Fusion of 16-bit GeoTIFF rasters, NetCDF-4/GRIIB2 atmospheric tensors, vector road networks (OSM), and relational CONAFOR fire logs.	- Unified harmonization layer (raster reprojection to UTM) - Common spatial denominator: regular 20m x 20m voxels - Conversion into unified columnar Apache Parquet matrices
VERACITY Atmospheric & Label Noise	Sensor distortions, cloud obstructions, & logging errors Dense cloud and shadow masks obscure optical reflectance. Historical fire registries often feature coordinate drift and reporting delays.	- Sentinel-2 Scene Classification Layer (SCL) bitmask filtering - Sentinel-1 SAR cross-polarization radar backscatter backup - Cross-referencing CONAFOR logs with NASA FIRMS thermal points
VALUE Tactical 48-72h Lead Time	Actionable hazard forecasting for disconnected frontline field units Transforms terabytes of planetary noise into 10m predictive risk tiles and plain-language feature attribution insights (TreeSHAP).	- High return on investment: zero telemetry hardware CapEx 48-72 hour operational window to preposition civil defense - Lightweight vector tile caching (PBF) in offline Progressive Web App

5. Data Ecosystem

TERRA-FIRE: DATA GENERATION & PROCESSING ECOSYSTEM

Phase 3 (Step 8) — High-throughput flow from raw observation repositories to machine learning and offline edge PWA

1. PLANETARY DATA GENERATION & EXTERNAL INGESTION

SATELLITE MISSIONS

Copernicus Sentinel-2 & 1

- Multispectral (B2-B12) 10-20m BOA
- Cadence: 5 Days | GEE & AWS STAC API

CLIMATE REANALYSIS

ECMWF ERA5-Land Reanalysis

- 2m Air Temp, Dewpoint, Soil Moisture, Wind
- Cadence: Hourly | Grid: 0.1° via CDS API

TERRAIN & TRUTH LOGS

Copernicus DEM & CONAFOR

- 30m GLO-30 Elevation, Slope & Aspect
- Static Topo + Historic Ignition Shapefiles

2. CLOUD PREPROCESSING, CALIBRATION & TOPOGRAPHIC DOWNSCALING

Optical QA & SAR Fusion

- SCL Bitmask: Cloud / Shadow Filtering
- Sentinel-1 Cross-Ratio (VH/VV) Fusion
- Mitigates optical occlusion & gaps

Microclimate Downscaling

- DEM Topographic Lapse-Rate (-6.5°C/km)
- Tetens Formulation for Vapor Pressure Deficit
- Resolves coarse 9km ERA5 down to 30m

Spatial Voxel Alignment

- Reprojection to Uniform UTM Metric Mesh
- Regular 20m x 20m Voxel Sampling
- Eliminates raster distortion and misalignment

3. SPATIOTEMPORAL FEATURE STORE & PREDICTIVE MACHINE LEARNING ENGINE

UNIFIED FEATURE STORE

Tabular Matrix (.parquet Arrays)

- Biophysical: NDMI, SAVI, NBR, Canopy Dehydration Velocity
- Thermodynamic: VPD (kPa), Soil Moisture L1, 10m Wind U/V
- Spatial: Slope (°), Aspect (Rad), Distance to Roads & Farms

PREDICTIVE ENGINE

XGBoost Classifier + TreeSHAP Kernel

- 48-to-72h Binary Ignition Prediction $P(\text{Fire} \in \{0, 1\})$
- Optimized for Extreme Class Imbalance (scale_pos_weight)
- Extracts Local Primary Risk Drivers for Field Explainability

4. TACTICAL EDGE DELIVERY & OFFLINE-FIRST PROGRESSIVE WEB APP (PWA)

VECTOR SERIALIZATION

Tippecanoe Tile Generation

- High-risk polygons clipped
- Compressed Mapbox Vector Tiles (.pbf)
- Compact JSON alert payloads

Payload: <15 MB per operational zone

EDGE CLIENT RUNTIME

PWA Storage Architecture

- Pre-caching in base via town Wi-Fi/LTE
- Service Worker background cache
- IndexedDB localized vector storage

Operates with 0% cellular connectivity

BRIGADE OPERATION

Tactical Field Decisions

- 48-72h Ignition Probability overlays
- Plain-language alerts: "Critical VPD"
- Safe canyon evacuation route planning

Low-memory MapLibre GL rendering



6. Variable Identification and Classification

VARIABLE INVENTORY & TAXONOMY: PRE-IGNITION HAZARD DOMAINS

Phase 4 (Step 9) — Comprehensive discovery of biophysical, climatic, topographic, and human risk predictors

BIOPHYSICAL FUEL

CANOPY & FUELBED STATE

NDMI (Normalized Moisture)

Sentinel-2 (B8 - B11)/(B8 + B11) • Live canopy water

SAVI (Soil-Adjusted Vigor)

Corrects background soil brightness in scrublands

NBR (Normalized Burn Ratio)

B8 vs B12 • Sensitive to charcoal & fuel desiccation

SAR Polarimetric Ratio (VH/VV)

Sentinel-1 C-SAR • Dielectric rough surface moisture

THERMODYNAMIC

ATMOSPHERIC EVAPORATIVE FORCE

VPD (Vapor Pressure Deficit)

Derived in kPa • Atmospheric suction drying fine fuels

Air Temperature (2m)

ERA5 downscaled to 30m via DEM adiabatic lapse rate

Wind Speed & Vectors (U10, V10)

10m horizontal velocity vectors • Flame oxygenation

Volumetric Soil Moisture (0-7cm)

Layer 1 volumetric water content (m³/m³)

TOPOGRAPHIC

TERRAIN PHYSICS & RELIEF

Slope Inclination (Degrees)

0° to 90° • Exponential flame convective pre-heating

Aspect Orientation (Radians)

Compass exposure • Solar radiation accumulation

Elevation (Meters ASL)

Copernicus GLO-30 • Bioclimatic vertical stratification

Topographic Wetness Index (TWI)

$\ln(a / \tan \beta)$ • Drainage accumulation proxy

HUMAN & TEMPORAL

ANTHROPOGENIC PRESSURES & SEASONALITY

Dist_to_Roads (Meters)

OpenStreetMap vectors • Proximate proxy for vehicle sparks & human access

Dist_to_Agriculture (Meters)

INEGI Land Use • Distance to slash-and-burn agricultural clearing borders

Day of Year (DOY) & Estiaje Month

Cyclical temporal encoding [sin/cos DOY] capturing dry season progression

Days Since Last Measurable Rain

Accumulated dry days from ERA5 precipitation time-series

SUPERVISED TARGET

GROUND-TRUTH DEPENDENT VARIABLES

Fire_Ignition_48h (Primary Target $\in \{0, 1\}$)

Binary flag: Confirmed combustion within 20m pixel voxel in the subsequent 48-72h
Derived from CONAFOR perimeters cross-referenced with VIIRS 375m thermal points

Burn_Severity_dNBR (Post-Fire Validation)

Differenced NBR (Pre-fire minus Post-fire) for damage validation

Historical Spatial Fire Frequency

Historical probability prior calculated over a 10-year CONAFOR baseline



VARIABLE CLASSIFICATION TABLE: STATISTICAL & MACHINE LEARNING TAXONOMY

Phase 4 (Step 10) — Formal mapping of predictive inputs, target labels, and mathematical scales

VARIABLE IDENTIFIER	INDEPENDENT	DEPENDENT	CATEGORICAL	NUMERICAL	MEASUREMENT UNIT / SCALE	PIPELINE SOURCE
Fire_Ignition_48h Primary ML Target Label	—	✓	✓	—	Binary Flag ∈ {0, 1}	CONAFOR / NASA FIRMS
NDMI Normalized Moisture Index	✓	—	—	✓	Continuous Ratio [-1.0, 1.0]	Sentinel-2 MSI (B8/B11)
VPD_kPa Vapor Pressure Deficit	✓	—	—	✓	Continuous Float (kPa)	ERA5-Land (Derived)
Temp_2m Ambient Air Temperature	✓	—	—	✓	Degrees Celsius (°C)	ERA5-Land (Downscaled)
Wind_Speed_10m Horizontal Velocity Magnitude	✓	—	—	✓	Meters per second (m/s)	ERA5-Land (U10, V10)
Soil_Moist_L1 Volumetric Soil Water (0-7cm)	✓	—	—	✓	m³ / m³ (0.0 to 0.60)	ERA5-Land Layer 1
Slope_Deg Terrain Inclination	✓	—	—	✓	Degrees (0° to 90°)	Copernicus GLO-30 DEM
Aspect_Rad Compass Direction of Slope	✓	—	—	✓	Radians [0, 2π]	Copernicus GLO-30 DEM
Land_Cover_Type Forest & Scrub Biome Class	✓	—	✓	—	Nominal String / Integer ID	INEGI Series VII
Dist_Roads_m Distance to Transport Network	✓	—	—	✓	Continuous Meters (m)	OpenStreetMap Vectors
Dist_Agri_m Distance to Agricultural Parcel	✓	—	—	✓	Continuous Meters (m)	INEGI / CONABIO
Days_Since_Rain Cumulative Precipitation Deficit	✓	—	—	✓	Discrete Integer (Days)	ERA5 Precipitation Time-Series



VARIABLE RELEVANCE ASSESSMENT: FEATURE PRIORITIZATION MATRIX

Phase 4 (Step 11) — Systematic ranking of predictive importance, biophysical rationale, and operational viability

VARIABLE NAME	PRIORITY LEVEL	PREDICTIVE RATIONALE & COMBUSTION PHYSICS	DATA QUALITY & PIPELINE FEASIBILITY IMPACT
VPD_kPa Vapor Pressure Deficit	CRITICAL (P1)	Governs the atmospheric drying suction on both fine dead fuels and living canopies. Direct physical ignition trigger.	Derived continuously from hourly ERA5-Land; requires DEM lapse-rate downscaling to 30m.
NDMI Canopy Moisture	CRITICAL (P1)	Strong linear correlation with Live Fuel Moisture Content (LFMC, $R^2 > 0.82$). Directly detects canopy wilting pre-fire.	10-20m resolution from Sentinel-2; requires cloud masking and SAR radar interpolation during overcast.
Fire_Ignition_48h Supervised Target Label	CRITICAL (P1)	Foundational binary ground truth necessary to train tree-based gradient boosting classifiers (XGBoost) over 48-72h windows.	Severe class imbalance (<0.05% positive pixels); requires scale_pos_weight & PR-AUC optimization.
Slope & Aspect Topographic Relief	HIGH (P2)	Flames pre-heat uphill vegetation exponentially faster; south/west aspects receive peak cumulative solar desiccation.	Static 30m Copernicus GLO-30 DEM; zero run-time latency, 100% cloud-free availability.
Soil_Moist_L1 Understory Wetness	HIGH (P2)	Indicates ground surface litter flammability. Critical baseline to prevent ground-to-canopy ladder fire transitions.	ERA5 Layer 1 volumetric index; robust availability but smoothed at macro-scale.
Wind_Speed_10m 10m Velocity Vectors	HIGH (P2)	Supplies oxygen to active ignition sites and governs potential rate of spread and ember spotting distances.	Hourly updates; requires canyon wind-tunnel proxies to capture mountain ridge acceleration.
Dist_Roads & Agri Human Interface Proxies	CONTEXTUAL (P3)	>85% of Mexican wildfires stem from escaped agricultural burns (roza y quema) and transit corridors.	OpenStreetMap & INEGI vectors; static Euclidean distance transforms computed during ETL.

7. Data Opportunity Map



TERRA-FIRE: COMPREHENSIVE DATA OPPORTUNITY MAP

Phase 5 (Step 12) — End-to-end traceability: Problem → Stakeholders → Decisions → Information Needs → Sources → Variables

1. CORE PROBLEM POST-IGNITION SUPPRESSION LAG & FRONTLINE INFORMATION BLINDNESS

National wildfire operations rely heavily on reactive thermal anomaly alerts (NASA FIRMS) and coarse weather grids (FW/SPPIF). Severe scarcity of rural mountain weather stations (<1 per 2,500 km²) and zero cellular connectivity prevent pre-ignition mitigation.

2. STAKEHOLDERS PRIMARY OPERATIONAL ACTORS & DECISION-MAKERS

Civil Protection Command

State & Municipal Coordinators (Logistics & Staging)

CONAFOR Regional Chiefs

Incident commanders & centralized fireline dispatchers

Ejidal & Volunteer Brigades

Frontline rural combatientes operating in zero-coverage zones

3. DECISIONS OPERATIONAL INTERVENTIONS (48-72H LEAD TIME)

Preemptive Asset Deployment

Pre-stage water tenders & crews in predicted critical zones

Agricultural Burn Moratoriums

Issue local bans on roza-y-quema when VPD & desiccation surge

Tactical Escape Navigation

Safe ingress/egress planning offline inside mountainous ravines

4. INFORMATION NEEDS REQUIRED BIOPHYSICAL & MICROCLIMATIC INTELLIGENCE

10–20m Live Fuel Moisture (LFMC)

Dynamic canopy water stress & physiological desiccation

Vapor Pressure Deficit (VPD)

Atmospheric drying power downscaled to 30m terrain grid

Offline Ignition Risk Surfaces

Vector tiles & explainable alert drivers (TreeSHAP indicators)

5. DATA SOURCES PLANETARY CONSTELLATIONS, REANALYSIS & FEDERAL REPOSITORIES

Sentinel-2 & 1 (ESA)

- B2–B12 Optical BOA (10–20m)
- C-SAR Radar Backscatter (VV/VH)

ERA5-Land (ECMWF)

- Hourly 2m Temp, Dewpoint, Wind
- Multi-layer Soil Wetness (0–7cm)

Copernicus GLO-30 DEM

- 30m Spatial Elevation Grid
- Topographic Slope, Aspect & TWI

CONAFOR & FIRMS

- Historical Fire Perimeters & Logs
- VIIRS 375m Thermal Hotspots

6. PREDICTIVE FEATURES ENGINEERED FEATURE MATRIX (STANDARDIZED 20M X 20M VOXELS IN PARQUET)

Target Variable (Dependent): Fire_Ignition_48h (Binary classification $\in \{0, 1\}$) • Calibrated for severe class imbalance via scale_pos_weight

Biophysical & Thermodynamic Features

- NDMI: $(B8 - B11)/(B8 + B11)$ Live Fuel Moisture Content proxy
- SAVI: Soil-adjusted vegetation index correcting arid scrubland
- VPD_kPa: True atmospheric drying suction force on forest fuel
- Soil_Moist_L1: 0–7cm understory litter drying baseline

Topographic & Anthropogenic Predictors

- Slope_Deg: Convective flame pre-heating uphill multiplier
- Aspect_Rad: Cumulative solar radiation desiccation orientation
- Dist_Roads_m: Proximity to transport corridors (spark catalysts)
- Dist_Agri_m: Euclidean distance to agricultural clearing burns



8. Gap Analysis

GAP ANALYSIS REPORT: TECHNICAL & ENVIRONMENTAL DATA LIMITATIONS

Phase 5 (Step 13) — Discrepancies between theoretical modeling needs, available data streams, and engineering mitigations

INFORMATION DOMAIN	REQUIRED OPERATIONAL STATE	CURRENT AVAILABLE STATE	IDENTIFIED TECHNICAL GAP	ENGINEERING MITIGATION STRATEGY
In-Situ Fuel Moisture Biophysical LFMC Calibration targets	Continuous in-situ LFMC sensor networks tracking canopy water content (g water / g dry matter)	Sparse weather stations across Mexico (<1 station per 2,500 km ² in mountains); minimal dedicated probes	CALIBRATION GAP Lack of continuous physical field targets to tune spectral reflectance ratios	- Implement biophysical transfer equations relating NDMI to LFMC ($R^2 > 0.82$) validated across pine-oak - Stratify by regional bioclimatic zones
Optical Satellite Coverage Sentinel-2 Multispectral Canopy desiccation tracking	Uninterrupted 5-day surface reflectance series during critical seasonal weather transitions	Sentinel-2 optical imagery frequently obstructed by dense cloud and cirrus fronts during summer/monsoon cycles	OPTICAL LATENCY GAP Missing reflectance values prior to dry spells when overcast covers mountain tops	- Fuse Sentinel-1 C-Band SAR radar cross-ratio (VH/VV) backscatter - Apply mathematical decay spline curves driven by cumulative local VPD
Micro-Meteorology ECMWF ERA5-Land VPD, wind, and temp	Granular 30m spatial resolution resolving valley thermal inversions and ridge wind speed	Hourly reanalysis provided at a coarse ~9 km grid (0.1° resolution) from Copernicus Climate Data Store	SPATIAL SCALE GAP Coarse pixels smooth sharp topographic elevation and ravine microclimates	- Topographic downscaling with DEM using adiabatic lapse rates (-6.5°C/km) - Model solar radiation aspect weights - Recalculate VPD at local 30m grid
Wildfire Ignition Labels CONAFOR & FIRMS Historical training target	Exact timestamp of spark and validated boundary polygon perimeters for supervised classification	CONAFOR registries record the date the fire was reported, not ignited; single point centroids stored	LABEL FIDELITY GAP Temporal discrepancies risk data leakage and false boundary training signals	- Cross-reference CONAFOR records with NASA FIRMS VIIRS 375m thermal points - Pinpoint first orbital thermal anomaly - Reconstruct perimeters via post-fire dNBR
Synthesis: Technical gaps are mitigated purely through mathematical downscaling, radar fusion, and cross-dataset conflation without hardware CapEx.				



9. Data Risk Assessment

DATA RISK MATRIX: TERRA-FIRE ACQUISITION & MACHINE LEARNING VULNERABILITIES

Phase 5 (Step 14) — Systematic evaluation of data quality, missing records, infrastructure constraints, and mitigation protocols

RISK ID & EVENT	SEVERITY	PROBABILITY	RISK LEVEL	OPERATIONAL & ANALYTICAL IMPACT	ENGINEERING COUNTERMEASURE & PROTOCOL
R-01: Severe Class Imbalance Target sparsity Ignition <0.05% voxels	High	High	CRITICAL	Model achieves 99.95% naive accuracy by defaulting to "No Fire", yielding 100% false negatives in field	- Implement scale_pos_weight in XGBoost - Discard accuracy; evaluate via PR-AUC - Apply spatial downsampling on negative voxels
R-02: Cloud Occlusion Optical data gaps Monsoon & cirrus fronts	High	Medium	HIGH	Missing optical reflectance bands during key transitional dry spells, stalling NDMI/SAVI canopy calculations	- C-SAR radar (Sentinel-1) cross-polarization - Mathematical spline decay curves tied to VPD - Automated fallback to pure meteorological pipeline
R-03: Spatial Leakage Data leakage Random k-fold validation	High	High	HIGH	Artificially optimistic cross-validation metrics that collapse when deployed to unseen geographical cordilleras	- Strictly enforce Blocked Spatial Cross-Validation - Leave-One-Region-Out cross-testing across biomes - Rolling temporal train/test validation splits
R-04: Mobile RAM Crashes Hardware constraints Low-tier Android devices	High	Medium	HIGH	Browser tabs crash in field conditions when parsing multi-megabyte GeoJSON layers inside offline WebGL canvases	- Compress polygons into Mapbox Vector Tiles (.pbf) - Dynamic Level-of-Detail (LoD) map clustering - Cap regional offline download payloads at <15 MB
R-05: Label Noise & Drift Incomplete records CONAFOR single points	Medium	Medium	MEDIUM	Centroid coordinates misalign with actual initial spark points; reporting dates cause temporal data leakage	- Cross-reference with NASA FIRMS VIIRS 375m - Anchor ignition to first orbital thermal detection - Reconstruct burn perimeters via post-fire dNBR
R-06: Upstream API Limits Data ownership & access Copernicus / CDS quotas	Medium	Low	MEDIUM	Scheduled daily ETL fails to download Sentinel-2 or ERA5-Land slices, delaying the morning 06:00 operational forecast	- Establish redundant cloud mirrors (AWS / GEE) - Localized rolling temporal caching of last 72h - Automated fallback to Open-Meteo REST APIs
Risk Protocol: All critical vulnerabilities (R-01 through R-04) are addressed via algorithmic loss functions, spatial cross-validation, and client-side vector tile compression.					

10. Preliminary Dataset Design

PRELIMINARY DATASET DESIGN: `terra\_fire\_spatiotemporal\_matrix\_v1.parquet`

Phase 6 (Step 16) — Unified 20m x 20m spatiotemporal feature store schema for supervised XGBoost ignition modeling

FEATURE COLUMN	DATA TYPE	PHYSICAL SOURCE	COLLECTION / INGEST	UPDATE CADENCE	OPERATIONAL & ANALYTICAL PURPOSE
voxel_id	String (UUID)	Spatial Mesh System	STATIC GRID	Permanent Key	Unique spatial hash anchor (20m x 20m)
fire_ignition_48h	Int8 ∈ {0, 1}	CONAFOR / NASA FIRMS	TARGET LABEL	48-72h Horizon	Supervised binary ground truth for XGBoost
ndmi_canopy_water	Float32 [-1, 1]	Sentinel-2 MSI (B8, B11)	5-DAY CYCLE	Rolling 5-Day	Live Fuel Moisture Content (LFMC) proxy
savi_vegetative_vigor	Float32 [-1, 1]	Sentinel-2 MSI (B4, B8)	5-DAY CYCLE	Rolling 5-Day	Understory vigor correcting soil reflectance
nbr_fuel_dryness	Float32 [-1, 1]	Sentinel-2 MSI (B8, B12)	5-DAY CYCLE	Rolling 5-Day	Combustion index & severe canopy drying
vpd_kpa	Float32 [kPa]	ERA5-Land (Downscaled)	HOURLY CDS	Daily (02:00 UTC)	Atmospheric evaporative suction pressure
temp_2m_celsius	Float32 [°C]	ERA5-Land (Lapse DEM)	HOURLY CDS	Daily (02:00 UTC)	Downscaled ambient temperature surface
wind_speed_ms	Float32 [m/s]	ERA5-Land (U10, V10)	HOURLY CDS	Daily (02:00 UTC)	Horizontal surface wind gust magnitude
soil_water_layer1	Float32 [m³/m³]	ERA5-Land (0-7cm)	DAILY CDS	Daily (02:00 UTC)	Ground litterbed moisture baseline
slope_degrees	Float32 [0°-90°]	Copernicus GLO-30 DEM	STATIC TOPO	Single Ingest	Flame pre-heating convective slope factor
aspect_radians	Float32 [0, 2π]	Copernicus GLO-30 DEM	STATIC TOPO	Single Ingest	Solar irradiation accumulation exposure
dist_roads_meters	Float32 [m]	OpenStreetMap Lines	SEMI-STATIC	Monthly Pull	Anthropogenic vehicle spark ignition prior
dist_agri_meters	Float32 [m]	INEGI / CONABIO	SEMI-STATIC	Seasonal Pull	Proximity to crop clearing burn borders
days_since_rain	Int16 [Days]	ERA5 Total Precipitation	DAILY CDS	Daily (02:00 UTC)	Cumulative drought desiccation tracker

Storage Architecture: Serialized as compressed Apache Parquet (Snappy) partitioned by `year=YYYY/month=MM/zone=utm=XX` for high-throughput distributed training.

Justification of Dataset Inclusions

- **Combined Spectral Moisture (ndmi_val, nbr_val):** Captures both canopy moisture and dead fuel desiccation dynamics, providing direct proxies for live fuel flammability.
- **Thermodynamic Predictors (vpd_kpa, soil_water_l1):** Represents the atmospheric evaporative demand pulling moisture out of biomass regardless of canopy color.
- **Topographical Predictors (slope_degrees, aspect_rad):** Encodes the physical mechanics of flame spread and micro-scale solar radiation accumulation.
- **Anthropogenic Distances (dist_road_m, dist_agri_m):** Accounts for human ignition catalysts, which cause the vast majority of wildfires in Mexico.



11. Strategic Recommendations

- **Cloud-Native Storage & Pipeline Automation:** Transition data collection scripts into containerized, serverless functions (e.g., Python on AWS Lambda or Google Cloud Functions) managed via Apache Airflow. Intermediate raster calculations should leverage Cloud-Optimized GeoTIFFs (COGs) and Zarr arrays to minimize redundant disk I/O.
- **Standardized Spatial Resolution & Gridding:** Adopt Uber's **H3 Discrete Global Grid System** (Resolution 9, approximately 100-meter hexagon edge length) or a regular $20\text{ m} \times 20\text{ m}$ UTM metric mesh as the common spatial denominator. This eliminates repeated resampling during model inference.
- **Automated Continuous Cloud Masking:** Implement automated cloud removal utilizing the Sentinel-2 Scene Classification Layer (SCL). Pixels flagged as clouds, cirrus, or cloud shadows must be temporally interpolated using trailing moving-average filters or SAR cross-polarization backscatter.
- **Spatial Cross-Validation Protocols:** Avoid traditional random k-fold cross-validation, which yields artificially inflated accuracy metrics due to spatial autocorrelation. Implement **Spatial Block Cross-Validation** or **Leave-One-Region-Out** cross-validation across distinct Mexican forest ecoregions (e.g., Sierra Madre Occidental vs. Sierra Madre del Sur).
- **Decoupled Client-Side Vector Caching:** Compress trained machine learning inferences into Mapbox Vector Tiles (MVT / [.pbf](#)). Store these tiles locally inside the mobile browser runtime using **IndexedDB**, ensuring field operators can interrogate hazard boundaries without internet connectivity.

12. Results Summary

This data opportunity mapping exercise yielded four key architectural and analytical insights:

- **The Pre-Ignition Window is Quantifiable:** Wildfires are not sudden anomalies; they are preceded by days of steady vegetative moisture depletion. Coupling Sentinel-2 short-wave infrared bands with ERA5-Land thermodynamic evaporative demand metrics creates a viable pre-ignition feature space.
- **Open Planetary Data Eliminates Hardware CapEx:** Expensive physical telemetry networks can be replaced by mature, open-access public data streams (Copernicus and ECMWF), providing an accessible, cost-effective data foundation.
- **Big Data Orchestration is Required:** The volume and variety of multi-band rasters and reanalysis grids demand automated cloud extract-transform-load (ETL) pipelines to reduce spatial data into clean, tabular Parquet feature stores.
- **Predictive Value Depends on Field Accessibility:** Highly accurate cloud models fail if their outputs cannot reach disconnected rural operators. The data strategy must prioritize lightweight, offline-first vector delivery via modern Progressive Web Application standards.



13. Challenges and Observations

- **Spatiotemporal Resolution Mismatch:** Reconciling ERA5-Land's coarse 9 km atmospheric cells with Sentinel-2's fine 10-meter optical grid presented an initial conceptual friction. This was resolved by using Digital Elevation Models (DEMs) to downscale temperature and vapor pressure fields based on localized adiabatic lapse rates and slope aspect.
- **Extreme Class Imbalance in Historical Data:** Active fire ignitions occur in a tiny fraction of spatial pixels across Mexico. Addressing this reality requires moving away from standard classification accuracy metrics toward Area Under the Precision-Recall Curve (AUPRC) and applying spatial negative down-sampling during training.
- **Historical Registry Gaps:** CONAFOR historical wildfire databases frequently record single point coordinates rather than precise perimeter shapes. Cross-referencing these institutional registries against NASA FIRMS thermal anomaly time-series provides a path to reconstruct accurate polygon perimeters.

14. Final Reflection

Reflection Report

Developing a deep understanding of data availability, structural variety, and underlying physical properties is an indispensable prerequisite to training analytical models or machine learning algorithms. In wildfire predictive modeling, jumping directly into model training without evaluating the data ecosystem invites severe systemic failures: models overfit to cloud artifacts, produce misleading evaluations due to spatial autocorrelation, or fail in the field because operators lack internet connectivity to query the model.

Mapping data opportunities establishes a rigorous operational baseline:

- It ensures every engineered feature corresponds directly to an established biophysical or thermodynamic mechanism (e.g., NDMI tracking leaf water content, VPD driving atmospheric drying).
- It forces data scientists to identify operational failure modes early—such as cloud occlusions or low-bandwidth rural conditions—and design engineering mitigations into the architecture.
- It guarantees that the resulting artificial intelligence application directly serves real-world operational decisions, providing actionable foresight rather than retrospective diagnostics.

15. Conclusions

The **TERRA-FIRE Data Opportunity Mapping** establishes that building a 48-to-72-hour predictive wildfire risk platform using open-source planetary data is technically, computationally, and operationally viable.

By strategically synthesizing Copernicus Sentinel-2 multispectral imagery, ECMWF ERA5-Land thermodynamic variables, and topographic elevation models into an optimized gradient-boosted decision tree architecture (XGBoost), the project shifts wildfire management from reactive suppression to proactive risk mitigation. The resulting tabular dataset schema, paired with an offline-first PWA caching strategy, provides a clear roadmap for subsequent academic phases: data acquisition, ETL pipeline development, machine learning model validation, and field-ready tactical deployment.

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